AN ALGEBRAIC STACK HAS UNRAMIFIED DIAGONAL EXACTLY WHEN IT ADMITS A SURJECTIVE ETALE SCHEME PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. DM in the source denotes the diagonal property; Deligne–Mumford denotes the existence-of-etale-atlas definition. Their equivalence is the counted construction, so neither is assumed as a synonym in the prerequisite list.

Editorial note. The adjacent quasi-DM flat/quasi-finite presentation theorem is not counted as a second task here.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Algebraic stacks, residual gerbes, smooth and unramified morphisms, regular local rings and regular sequences, etale localization, openness and finite-type points. The local slicing and DM residual-gerbe arguments are retained author proof.

Source reference. Original lemma 04NG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6079–6079. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00LM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 792–792. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06LW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8480–8480. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06M2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8486–8486. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06MP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8508–8508. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 02ZY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3929–3929. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 04WY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6376–6376. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06FF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8249–8249. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06LZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8483–8483. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06LH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8466–8466. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05VH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7551–7551. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 03BM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4353–4353. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original equation 03BF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4346–4346. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06MM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8506–8506. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original theorem 04S6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6209–6209. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04S4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6207–6207. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 046T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5484–5484. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02L2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3393–3393. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02KY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3389–3389. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02VI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3769–3769. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02L3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3394–3394. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04R9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6177–6177. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02G5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3216–3216. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02GV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3242–3242. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04XD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6391–6391. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 03YO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5191–5191. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06MB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8495–8495. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06G3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8272–8272. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06MZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8518–8518. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03XG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5147–5147. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06LT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8477–8477. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04XH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6395–6395. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05W3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7572–7572. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05W5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7574–7574. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05W1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7570–7570. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03ZD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5216–5216. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03ZI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5221–5221. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 056U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6722–6722. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 056S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6720–6720. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02GF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3226–3226. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00NQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 868–868. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06N2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8521–8521. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04XL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6399–6399. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 06G2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8271–8271. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04XI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6396–6396. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Theorem 0.1. *Let $\mathcal{X}$ be an algebraic stack. The following are equivalent*

(1) *$\mathcal{X}$ is DM,*
(2) *$\mathcal{X}$ is Deligne-Mumford, and*
(3) *there exists a scheme W and a surjective étale morphism $W \to \mathcal{X}$.*

Lemma 0.2. *Let $\mathcal{X}$ be an algebraic stack. Consider a cartesian diagram*

$$\begin{array}{ccc} U & \xleftarrow{p} & F \\ \downarrow & & \downarrow \\ \mathcal{X} & \longleftarrow & \mathrm{Spec}(k) \end{array}$$

where U is an algebraic space, k is a field, and $U \to \mathcal{X}$ is flat and locally of finite presentation. Let $f_1, \ldots, f_r \in \Gamma(U, \mathcal{O}_U)$ and $z \in |F|$ such that $f_1, \ldots, f_r$ map to a regular sequence in the local ring $\mathcal{O}_{F,\overline{z}}$. Then, after replacing U by an open subspace containing $p(z)$, the morphism

$$V(f_1, \ldots, f_r) \longrightarrow \mathcal{X}$$

is flat and locally of finite presentation.

Proof. Choose a scheme W and a surjective smooth morphism $W \to \mathcal{X}$. Choose an extension of fields k'/k and a morphism $w : \mathrm{Spec}(k') \to W$ such that $\mathrm{Spec}(k') \to W \to \mathcal{X}$ is 2-isomorphic to $\mathrm{Spec}(k') \to \mathrm{Spec}(k) \to \mathcal{X}$. This is possible as $W \to \mathcal{X}$ is surjective. Consider the commutative diagram

$$\begin{array}{ccccc} U & \xleftarrow{\mathrm{pr}_0} & U \times_{\mathcal{X}} W & \xleftarrow{p'} & F' \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{X} & \longleftarrow & W & \longleftarrow & \mathrm{Spec}(k') \end{array}$$

both of whose squares are cartesian. By our choice of w we see that $F' = F \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k')$. Thus $F' \to F$ is surjective and we can choose a point $z' \in |F'|$ mapping to z. Since $F' \to F$ is flat we see that $\mathcal{O}_{F,\overline{z}} \to \mathcal{O}_{F',\overline{z}'}$ is flat, see Morphisms of Spaces, Lemma 04NG. Hence $f_1, \ldots, f_r$ map to a regular sequence in $\mathcal{O}_{F',\overline{z}'}$, see Algebra, Lemma 00LM. Note that $U \times_{\mathcal{X}} W \to W$ is a morphism of algebraic spaces which is flat and locally of finite presentation. Hence by More on Morphisms of Spaces, Lemma 06LW we see that there exists an open subspace U' of $U \times_{\mathcal{X}} W$ containing $p'(z')$ such that the intersection $U' \cap (V(f_1, \ldots, f_r) \times_{\mathcal{X}} W)$ is flat and locally of finite presentation over W. Note that $\mathrm{pr}_0(U')$ is an open subspace of U containing $p(z)$ as pr_0 is smooth hence open. Now we see that $U' \cap (V(f_1, \ldots, f_r) \times_{\mathcal{X}} W) \to \mathcal{X}$ is flat and locally of finite presentation as the composition

$$U' \cap (V(f_1, \ldots, f_r) \times_{\mathcal{X}} W) \to W \to \mathcal{X}.$$

Hence Properties of Stacks, Lemma 06M2 implies $\mathrm{pr}_0(U') \cap V(f_1, \ldots, f_r) \to \mathcal{X}$ is flat and locally of finite presentation as desired. □

Lemma 0.3. *Let $\mathcal{Z}$ be a DM, locally Noetherian, reduced algebraic stack with $|\mathcal{Z}|$ a singleton. Then there exists a field k and a surjective étale morphism $\mathrm{Spec}(k) \to \mathcal{Z}$.*

Proof. By Properties of Stacks, Lemma 06MP there exists a field k and a surjective, flat, locally finitely presented morphism $\mathrm{Spec}(k) \to \mathcal{Z}$. Set $U = \mathrm{Spec}(k)$ and $R = U \times_{\mathcal{Z}} U$ so we obtain a groupoid in algebraic spaces (U, R, s, t, c), see Algebraic Stacks, Lemma 02ZY. Note that by Algebraic Stacks, Remark 04WY we have an equivalence

$$f_{can} : [U/R] \longrightarrow \mathcal{Z}$$

The projections $s, t : R \to U$ are locally of finite presentation. As $\mathcal{Z}$ is DM we see that the stabilizer group algebraic space

$$G = U \times_{U \times U} R = U \times_{U \times U} (U \times_{\mathcal{Z}} U) = U \times_{\mathcal{Z} \times \mathcal{Z}, \Delta_{\mathcal{Z}}} \mathcal{Z}$$

is unramified over U. In particular $\dim(G) = 0$ and by More on Groupoids in Spaces, Lemma 06FF we have $\dim(R) = 0$. This implies that R is a scheme, see Spaces over Fields, Lemma 06LZ. By Varieties, Lemma 06LH we see that R (and also G) is the disjoint union of spectra of Artinian local rings finite over k via either s or t. Let $P = \mathrm{Spec}(A) \subset R$ be the open and closed subscheme whose underlying point is the identity e of the groupoid scheme (U, R, s, t, c). As $s \circ e = t \circ e = \mathrm{id}_{\mathrm{Spec}(k)}$ we see that A is an Artinian local ring whose residue field is identified with k via either $s^\sharp : k \to A$ or $t^\sharp : k \to A$. Note that $s, t : \mathrm{Spec}(A) \to \mathrm{Spec}(k)$ are finite (by the lemma referenced above). Since $G \to \mathrm{Spec}(k)$ is unramified we see that

$$G \cap P = P \times_{U \times U} U = \mathrm{Spec}(A \otimes_{k \otimes k} k)$$

is unramified over k. On the other hand $A \otimes_{k \otimes k} k$ is local as a quotient of A and surjects onto k. We conclude that $A \otimes_{k \otimes k} k = k$. It follows that $P \to U \times U$ is universally injective (as P has only one point with residue field k), unramified (by the computation of the fibre over the unique image point above), and of finite type (because s, t are) hence a monomorphism (see Étale Morphisms, Lemma 05VH). Thus $s|_P, t|_P : P \to U$ define a finite flat equivalence relation. Thus we may apply Groupoids, Proposition 03BM to conclude that U/P exists and is a scheme $\overline{U}$. Moreover, $U \to \overline{U}$ is finite locally free and $P = U \times_{\overline{U}} U$. In fact $\overline{U} = \mathrm{Spec}(k_0)$ where $k_0 \subset k$ is the ring of R-invariant functions. As k is a field it follows from the definition Groupoids, Equation (03BF) that k_0 is a field.

We claim that

$$\mathrm{Spec}(k_0) = \overline{U} = U/P \to [U/R] = \mathcal{Z} \tag{0.3.1}$$

is the desired surjective étale morphism. It follows from Properties of Stacks, Lemma 06MM that this morphism is surjective. Thus it suffices to show that (0.3.1) is étale[1]. Instead of proving the étaleness directly we first apply Bootstrap, Lemma 04S4 to see that there exists a groupoid scheme $(\overline{U}, \overline{R}, \overline{s}, \overline{t}, \overline{c})$ such that (U, R, s, t, c) is the restriction of $(\overline{U}, \overline{R}, \overline{s}, \overline{t}, \overline{c})$ via the quotient morphism $U \to \overline{U}$. (We verified all the hypothesis of the lemma above except for the assertion that $j : R \to U \times U$ is separated and locally quasi-finite which follows from the fact that R is a separated scheme locally quasi-finite over k.) Since $U \to \overline{U}$ is finite locally free we see that $[U/R] \to [\overline{U}/\overline{R}]$ is an equivalence, see Groupoids in Spaces, Lemma 046T.

Note that s, t are the base changes of the morphisms $\overline{s}, \overline{t}$ by $U \to \overline{U}$. As $\{U \to \overline{U}\}$ is an fppf covering we conclude $\overline{s}, \overline{t}$ are flat, locally of finite presentation, and

[1] We urge the reader to find his/her own proof of this fact. In fact the argument has a lot in common with the final argument of the proof of Bootstrap, Theorem 04S6 hence probably should be isolated into its own lemma somewhere.

locally quasi-finite, see Descent, Lemmas 02L2, 02KY, and 02VI. Consider the commutative diagram

$$\begin{array}{ccccc} U \times_{\overline{U}} U & = & P & \longrightarrow & R \\ & \searrow & \downarrow & & \downarrow \\ & & \overline{U} & \xrightarrow{\overline{e}} & \overline{R} \end{array}$$

It is a general fact about restrictions that the outer four corners form a cartesian diagram. By the equality we see the inner square is cartesian. Since P is open in R we conclude that $\overline{e}$ is an open immersion by Descent, Lemma 02L3.

But of course, if $\overline{e}$ is an open immersion and $\overline{s}, \overline{t}$ are flat and locally of finite presentation then the morphisms $\overline{t}, \overline{s}$ are étale. For example you can see this by applying More on Groupoids, Lemma 04R9 which shows that $\Omega_{\overline{R}/\overline{U}} = 0$ implies that $\overline{s}, \overline{t} : \overline{R} \to \overline{U}$ is unramified (see Morphisms, Lemma 02G5), which in turn implies that $\overline{s}, \overline{t}$ are étale (see Morphisms, Lemma 02GV). Hence $\mathcal{Z} = [\overline{U}/\overline{R}]$ is an étale presentation of the algebraic stack $\mathcal{Z}$ and we conclude that $\overline{U} \to \mathcal{Z}$ is étale by Properties of Stacks, Lemma 04XD. □

Proof. Recall that (3) is the definition of (2), see Algebraic Stacks, Definition 03YO. The implication (3) $\Rightarrow$ (1) is Lemma 06MB. Assume (1). Let $x \in |\mathcal{X}|$ be a finite type point. We will produce a scheme over $\mathcal{X}$ which "works" in a neighbourhood of x. At the end of the proof we will take the disjoint union of all of these to conclude.

By Lemma 06G3 the residual gerbe $\mathcal{Z}_x$ of $\mathcal{X}$ at x exists and $\mathcal{Z}_x \to \mathcal{X}$ is locally of finite type. By Lemma 06MZ the algebraic stack $\mathcal{Z}_x$ is DM. By Lemma 0.3 there exists a field k and a surjective étale morphism $z : \mathrm{Spec}(k) \to \mathcal{Z}_x$. In particular the composition $x : \mathrm{Spec}(k) \to \mathcal{X}$ is locally of finite type (by Morphisms of Spaces, Lemmas 03XG and 06LT).

Pick a scheme U and a smooth morphism $U \to \mathcal{X}$ such that x is in the image of $|U| \to |\mathcal{X}|$. Consider the following fibre square

$$\begin{array}{ccc} U & \longleftarrow & F \\ \downarrow & & \downarrow \\ \mathcal{X} & \xleftarrow{x} & \mathrm{Spec}(k) \end{array}$$

in other words $F = U \times_{\mathcal{X},x} \mathrm{Spec}(k)$. By Properties of Stacks, Lemma 04XH we see that F is nonempty. As $\mathcal{Z}_x \to \mathcal{X}$ is a monomorphism we have

$$\mathrm{Spec}(k) \times_{z,\mathcal{Z}_x,z} \mathrm{Spec}(k) = \mathrm{Spec}(k) \times_{x,\mathcal{X},x} \mathrm{Spec}(k)$$

with étale projection maps to $\mathrm{Spec}(k)$ by construction of z. Since

$$F \times_U F = (\mathrm{Spec}(k) \times_{\mathcal{X}} \mathrm{Spec}(k)) \times_{\mathrm{Spec}(k)} F$$

we see that the projections maps $F \times_U F \to F$ are étale as well. It follows that $\Delta_{F/U} : F \to F \times_U F$ is étale (see Morphisms of Spaces, Lemma 05W3). By Morphisms of Spaces, Lemma 05W5 this implies that $\Delta_{F/U}$ is an open immersion, which finally implies by Morphisms of Spaces, Lemma 05W1 that $F \to U$ is unramified.

Pick a nonempty affine scheme V and an étale morphism $V \to F$. (This could be avoided by working directly with F, but it seems easier to explain what's going on by doing so.) Picture

$$\begin{array}{ccccc} U & \longleftarrow & F & \longleftarrow & V \\ \downarrow & & \downarrow & \swarrow & \\ \mathcal{X} & \xleftarrow{x} & \mathrm{Spec}(k) & & \end{array}$$

Then $V \to \mathrm{Spec}(k)$ is a smooth morphism of schemes and $V \to U$ is an unramified morphism of schemes (see Morphisms of Spaces, Lemmas 03ZD and 03ZI). Pick a closed point $v \in V$ with $k \subset \kappa(v)$ finite separable, see Varieties, Lemma 056U. Let $u \in U$ be the image point. The local ring $\mathcal{O}_{V,v}$ is regular (see Varieties, Lemma 056S) and the local ring homomorphism

$$\varphi : \mathcal{O}_{U,u} \longrightarrow \mathcal{O}_{V,v}$$

coming from the morphism $V \to U$ is such that $\varphi(\mathfrak{m}_u)\mathcal{O}_{V,v} = \mathfrak{m}_v$, see Morphisms, Lemma 02GF. Hence we can find $f_1, \ldots, f_d \in \mathcal{O}_{U,u}$ such that the images $\varphi(f_1), \ldots, \varphi(f_d)$ form a basis for $\mathfrak{m}_v/\mathfrak{m}_v^2$ over $\kappa(v)$. Since $\mathcal{O}_{V,v}$ is a regular local ring this implies that $\varphi(f_1), \ldots, \varphi(f_d)$ form a regular sequence in $\mathcal{O}_{V,v}$ (see Algebra, Lemma 00NQ). After replacing U by an open neighbourhood of u we may assume $f_1, \ldots, f_d \in \Gamma(U, \mathcal{O}_U)$. After replacing U by a possibly even smaller open neighbourhood of u we may assume that $V(f_1, \ldots, f_d) \to \mathcal{X}$ is flat and locally of finite presentation, see Lemma 0.2. By construction

$$V(f_1, \ldots, f_d) \times_{\mathcal{X}} \mathrm{Spec}(k) \longleftarrow V(f_1, \ldots, f_d) \times_U V$$

is étale and $V(f_1, \ldots, f_d) \times_U V$ is the closed subscheme $T \subset V$ cut out by $f_1|_V, \ldots, f_d|_V$. Hence by construction $v \in T$ and

$$\mathcal{O}_{T,v} = \mathcal{O}_{V,v}/(\varphi(f_1), \ldots, \varphi(f_d)) = \kappa(v)$$

a finite separable extension of k. It follows that $T \to \mathrm{Spec}(k)$ is unramified at v, see Morphisms, Lemma 02GF. By definition of an unramified morphism of algebraic spaces this means that $V(f_1, \ldots, f_d) \times_{\mathcal{X}} \mathrm{Spec}(k) \to \mathrm{Spec}(k)$ is unramified at the image of v in $V(f_1, \ldots, f_d) \times_{\mathcal{X}} \mathrm{Spec}(k)$. Applying Lemma 06N2 we see that on shrinking U to yet another open neighbourhood of u the morphism $V(f_1, \ldots, f_d) \to \mathcal{X}$ is étale.

We conclude that for every finite type point x of $\mathcal{X}$ there exists an étale morphism $f_x : W_x \to \mathcal{X}$ with x in the image of $|f_x|$. Set $W = \coprod_x W_x$ and $f = \coprod f_x$. Then f is étale. In particular the image of $|f|$ is open, see Properties of Stacks, Lemma 04XL. By construction the image contains all finite type points of $\mathcal{X}$, hence f is surjective by Lemma 06G2 (and Properties of Stacks, Lemma 04XI). □

References